

Reinforcement Learning – Other Topics 3

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Topics..

- Tips and Tricks for Practical RL
- **Advanced Topics:**
 - Model-based Reinforcement Learning
 - Imitation Learning
 - RL in Language Models
 - Multi-Agent RL

Multi-Agent RL

Multi-Agent Reinforcement Learning

When we do multi-agents reinforcement learning (MARL), we are in a situation where we have **multiple agents that share and interact in a common environment**.



Multi-Agent Reinforcement Learning

Multi-agent environments are **often modeled as games**, even for real-world problems. Different game types are categorized based on reward structures:

- Zero-sum: $\sum_i R_i(a) = 0$ for all a (e.g., Rock-Paper-Scissors).
- Common-reward: $R_i(a) = R_j(a)$ for all i, j, a (e.g., Coordination Game).
- General-sum: No restrictions on rewards (e.g., Prisoner's Dilemma).

Stochastic Games

- Multi-agent version of an MDP. Defined by:
 - Finite set of agents I
 - Finite set of states S
 - Finite action sets A_i for each agent $i \in I$
 - Reward function $R_i(s, a, s')$
 - State transition distribution $T(s_{t+1}|s_{1:t}, a_t)$
 - Initial state distribution $\mu(s_0)$
- Policy for each agent: $\pi_i(a_t|h_t)$, where h_t is the state-action history.

Partially Observable Stochastic Games (POSGs)

- Multi-agent version of a POMDP.
- Augments stochastic games with observation distributions $O_i(o_{t+1}|s_{t+1}, a_t)$ for each agent i .
- Policy for each agent: $\pi_i(a_t|h_i)$, where h_i is the observation history for agent i .
- Dec-POMDP is a special case with a common reward function for all agents.

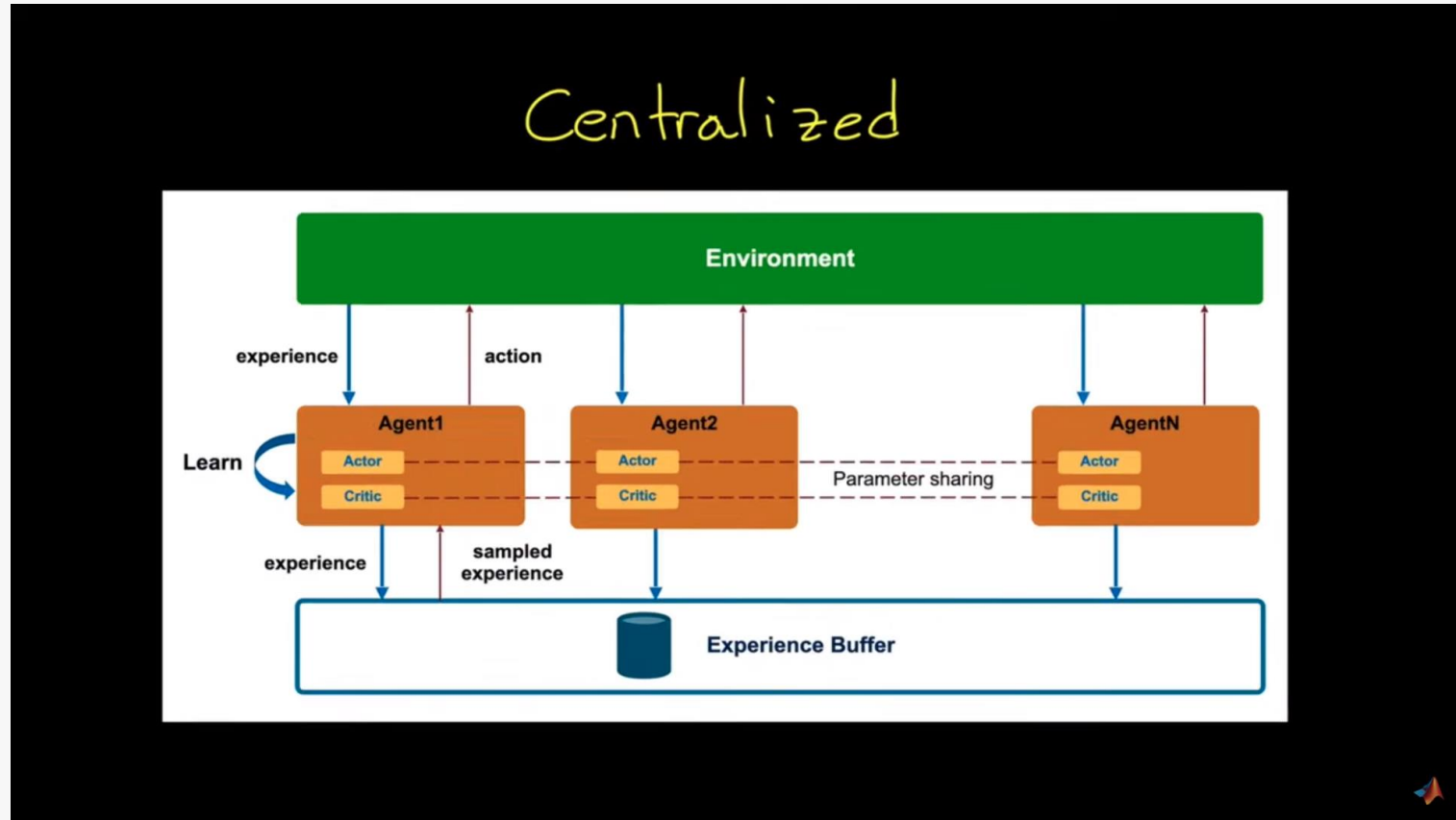
Reductions to Single-Agent Learning

Two approaches simplify MARL by reducing it to single-agent RL:

Centralized Learning: Learns a single joint policy; difficult for general-sum games and doesn't scale well.

Independent Learning: Each agent treats others as part of the environment; issues with non-stationarity.

High-Level Overview

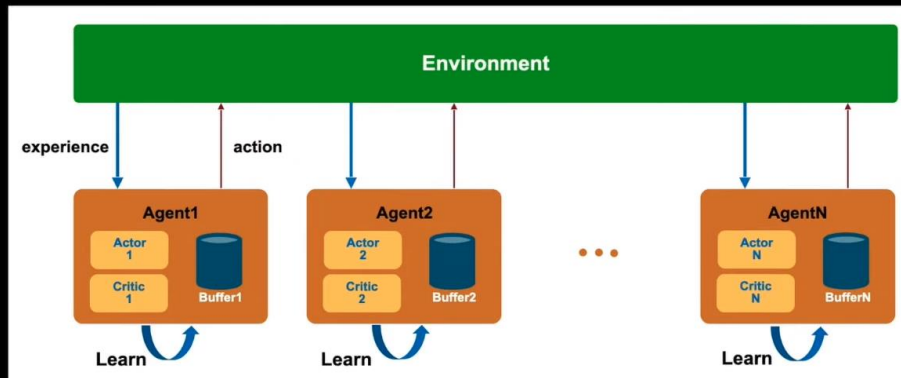


Source: [Introduction to Multi-Agent Reinforcement Learning](#)

High-Level Overview

MARL Approaches

Decentralized



Source: [Introduction to Multi-Agent Reinforcement Learning](#)

Kind of Environments

In a multi-agent system agents interact with other agents. we can have different types of environments:

- *Cooperative environments*: where your agents need **to maximize the common benefits**.

For instance, in a warehouse, **robots must collaborate to load and unload the packages efficiently (as fast as possible)**.

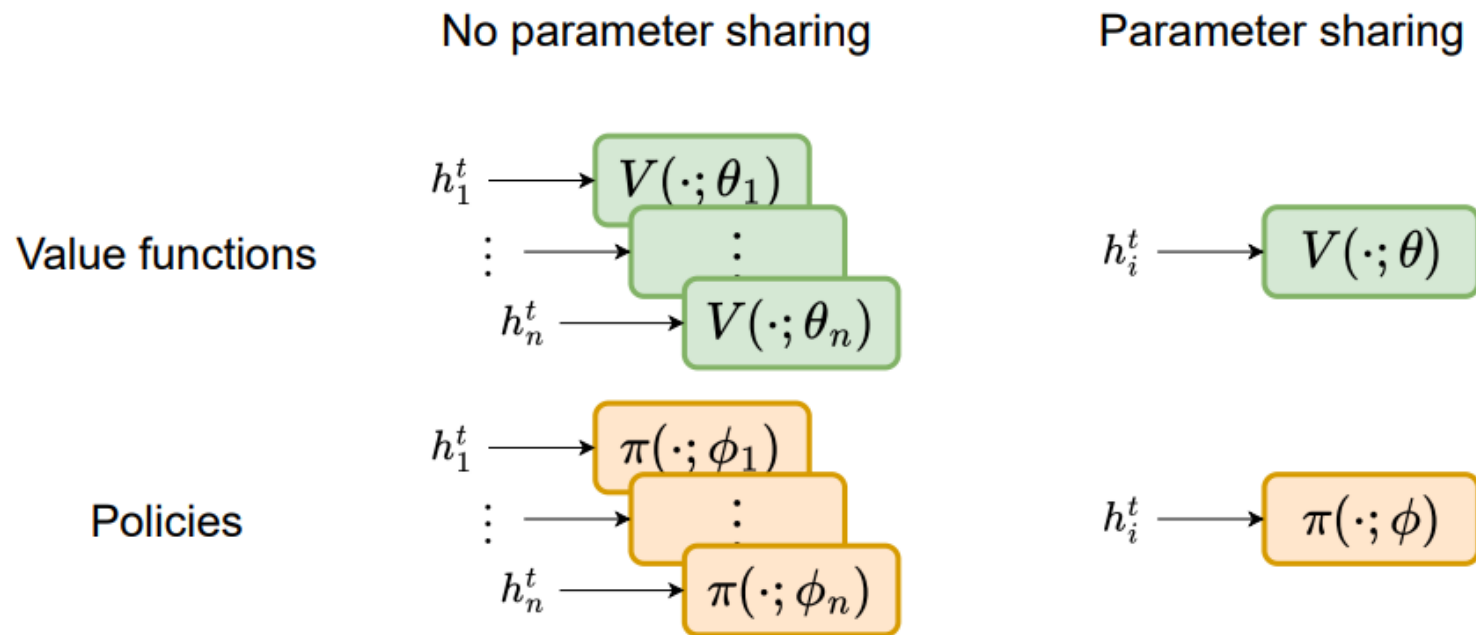
- *Competitive/Adversarial environments*: in this case, your agent **wants to maximize its benefits by minimizing the opponent's**.

For example, in a game of tennis, **each agent wants to beat the other agent**.

- *Mixed of both adversarial and cooperative*: like in a football environment, multiple agents are part of a team and they need to cooperate with each other and beat the opponent team.

Parameter Sharing in MARL

Sharing network parameters across agents is common practice to make MARL training more efficient. Share parameters across value functions, policies, or both.



Parameter Sharing in MARL

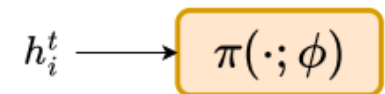
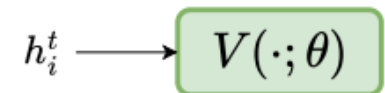
Sharing network parameters across agents is common practice to make MARL training more efficient. Share parameters across value functions, policies, or both.

Parameter sharing has two primary benefits:

- **Scalability:** the number of parameters remains constant independent of the number of agents $\rightarrow$ less computational cost
- **Efficiency:** shared parameters are updated using the experiences of all agents $\rightarrow$ more training data for the shared parameters

The downside is that (naive) parameter sharing assumes strongly homogeneous agents.

Parameter sharing



Extensions of PPO to Multi-Agent

- Independent PPO (IPPO):
 - Each agent is trained independently using PPO.
 - No explicit communication or central coordination during training.
 - Scalable but may struggle with non-stationarity and coordination.
- Multi-Agent PPO (MAPPO):
 - Centralized critic observes global state and all agents' actions.
 - Decentralized execution with policies conditioned only on local observations.
 - Better handles non-stationarity and coordination, at the cost of additional complexity.

Extensions of PPO to Multi-Agent

- Key Differences between IPPO and MAPPO:
 - Training: Independent vs Centralized critic
 - Complexity: Simple and scalable vs More complex but coordinated
 - Performance: May falter in cooperative tasks vs Stronger coordination

Communication-Based MARL

Why Communication in MARL?

- **Non-stationarity:** stabilize learning by sharing information between agents.
- **Coordination:** enable cooperative behaviors and joint strategies.
- **Partial Observability:** share local observations to improve world model.
- **Scalability:** structured communication reduces complexity in large teams.

Communication-Based MARL

1. Differentiable Communication Channels

- CommNet: learn continuous message vectors via backpropagation.

2. Discrete Communication

- DIAL: differentiable inter-agent learning with discrete messages.

3. Attention-Based & Graph Networks

- TarMAC: targeted multi-agent communication using soft attention.

Learned Models: Policies & Communication

- **Action Policy:**

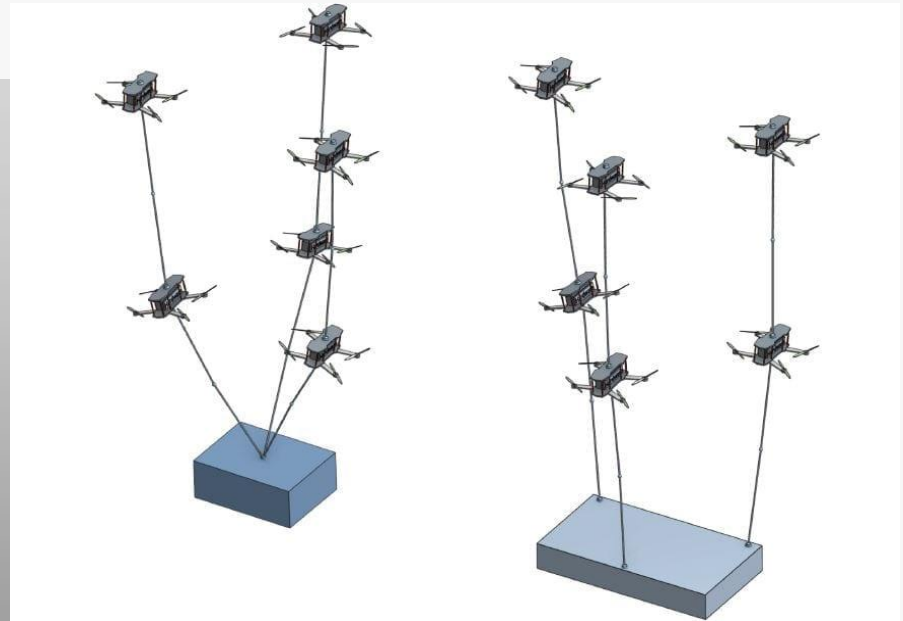
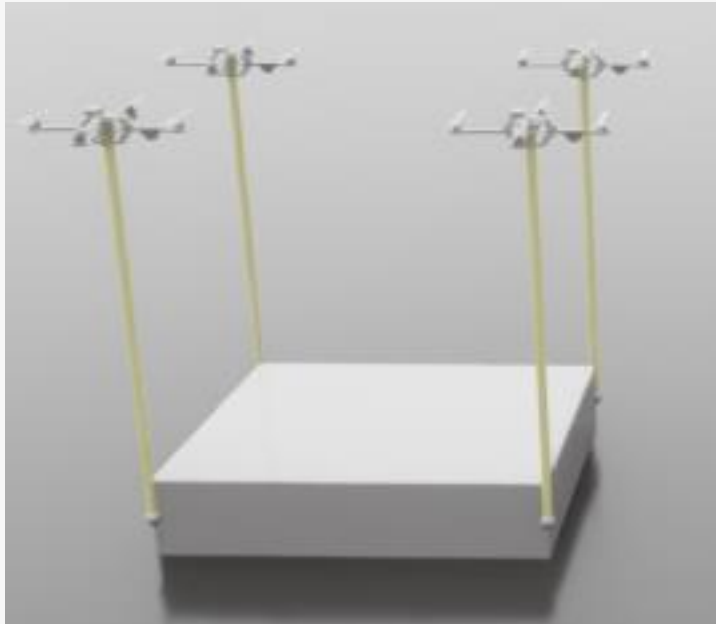
- $\pi_i^a(a_i^t \mid o_i^t, m_i^t; \theta_i)$
- where $m_i^t = f^c(h_i^t; \varphi_i)$ is the message generated

- **Communication Policy:**

- $\pi_i^c(m_i^t \mid o_i^t; \varphi_i)$

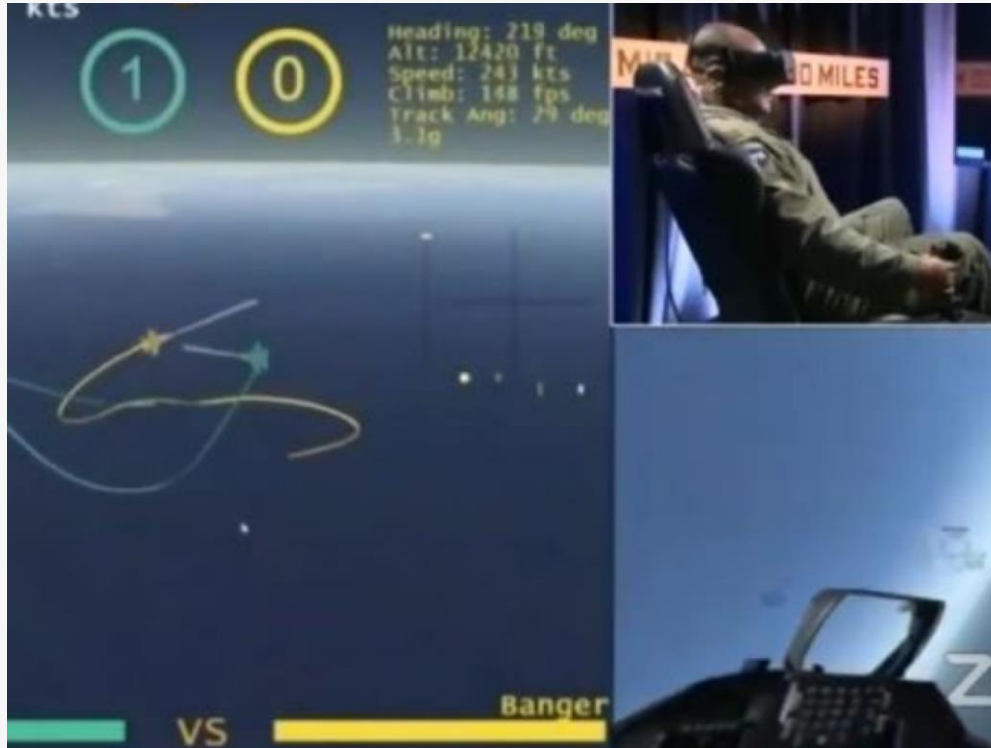
Applications to Engineering

- UAV Swarm Coordination



Applications to Engineering

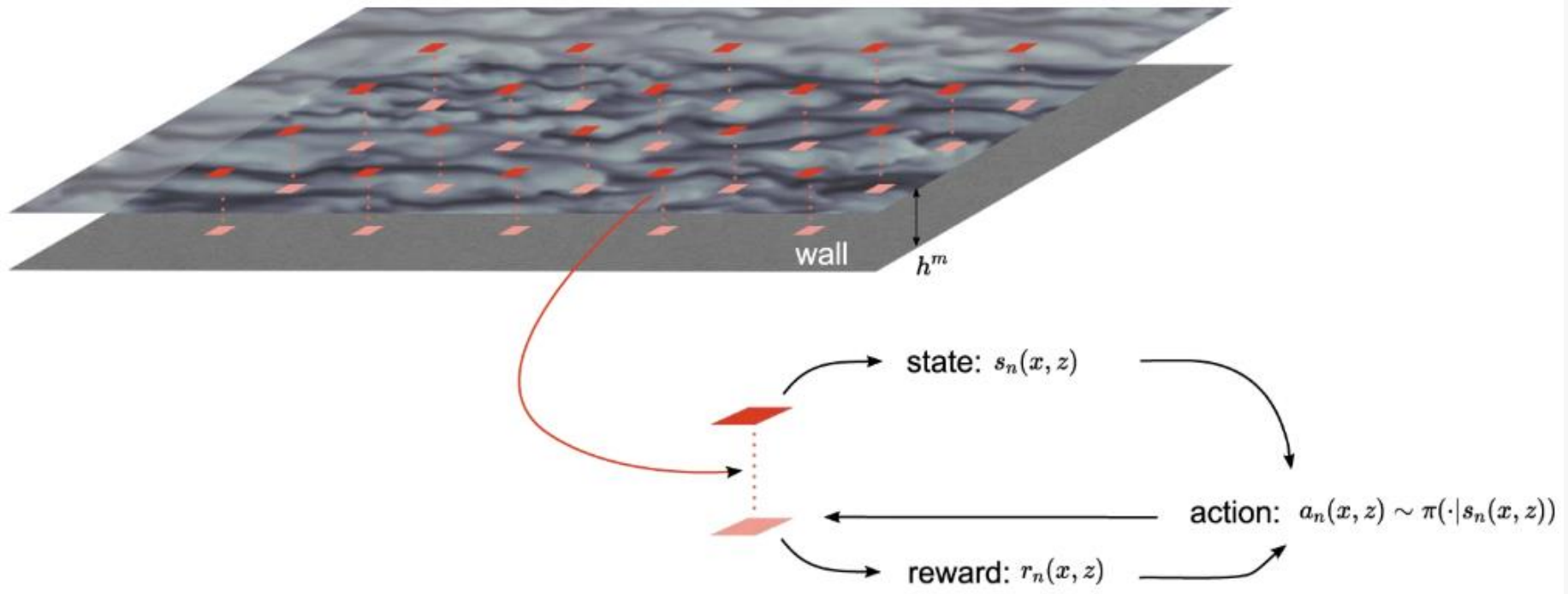
- Autonomous Dogfighting



[DARPA's AlphaDogfight Trial's Final Round: Human F-16 Pilot Lost Against AI For the First Time](#)

Applications to Engineering

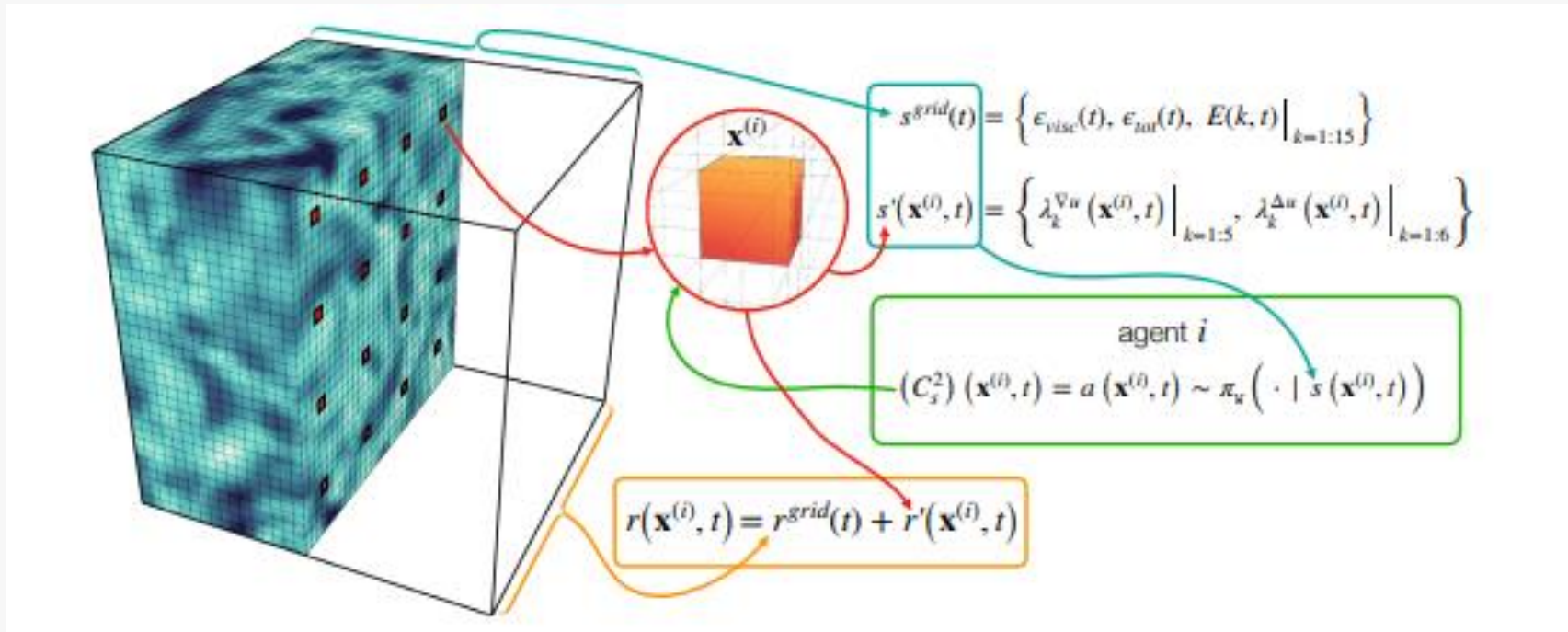
- Near-Wall Turbulence Modeling



Bae, H.J., Koumoutsakos, P. Scientific multi-agent reinforcement learning for wall-models of turbulent flows. *Nat Commun* **13**, 1443 (2022). <https://doi.org/10.1038/s41467-022-28957-7>

Applications to Engineering

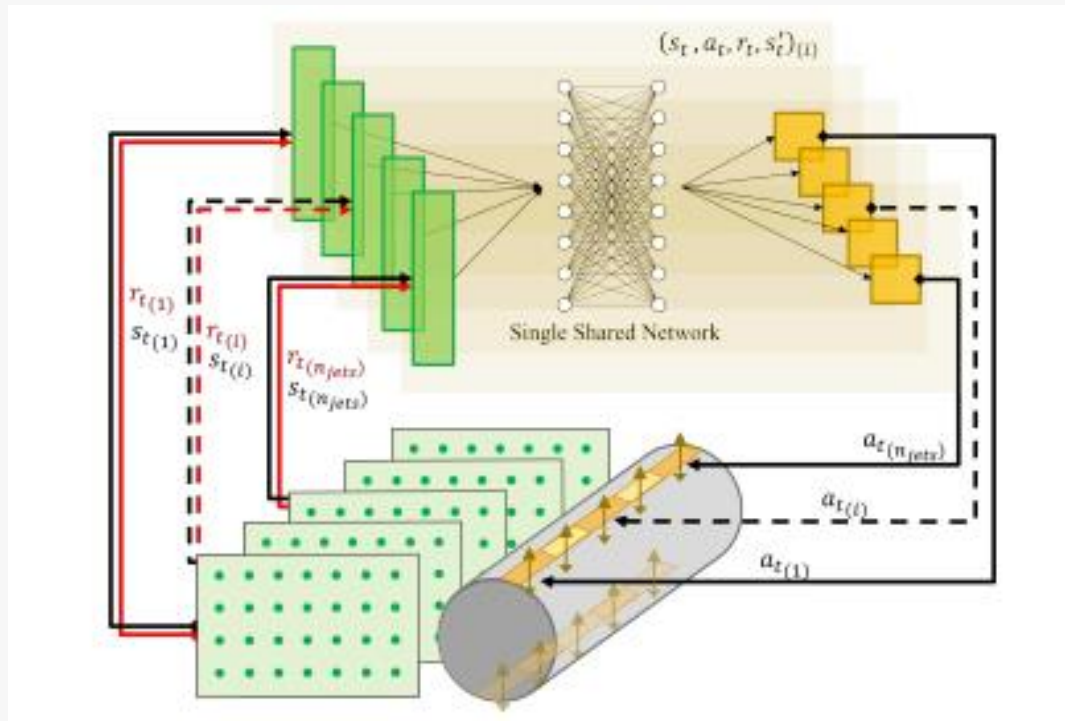
- Turbulence Modeling



Novati, G., de Laroussilhe, H.L. & Koumoutsakos, P. Automating turbulence modelling by multi-agent reinforcement learning. Nat Mach Intell 3, 87–96 (2021). <https://doi.org/10.1038/s42256-020-00272-0>

Applications to Engineering

- Complex Flow Control



Suárez, P., Álcantara-Ávila, F., Rabault, J., Miró, A., Font, B., Lehmkuhl, O., & Vinuesa, R. (2024). Flow control of three-dimensional cylinders transitioning to turbulence via multi-agent reinforcement learning. *arXiv preprint arXiv:2405.17210*.

Acknowledgments

- CS 285 at UC Berkeley, Deep Reinforcement Learning. Sergey Levine
- Reinforcement Learning: A Comprehensive Overview. Kevin P. Murphy
- Advanced Topics in Learning and Decision Making, Pieter Abbeel. UC Berkeley
- HuggingFace LLM course
- Multi-Agent Reinforcement Learning: Foundations and Modern Approaches by Stefano V. Albrecht, Filippos Christianos and Lukas Schäfer MIT Press, 2024
- HuggingFace Introduction to RL course